

ENGINEERING EXPLORATION



WHAT IS ENGINEERING:

Engineering is the application of scientific and mathematical principles to design, build, and improve things that solve practical problems — structures, machines, systems, processes, or software.



DIFFERENCE BETWEEN SCIENCE AND ENGINEERING



SCIENCE

VS

ENGINEERING

SCIENCE	ENGINEERING
• Discover knowledge	• Create solution
• Studies of natural world	• Engineering applies scientific knowledge to solve problems
• Ex:studing gravity or electricity	• Ex:Designing bridges,computers,or mobile apps
• It answers "why" and "how" things happens.	• It answers "how can we make or improve it"

DIFFERENCE BETWEEN SCIENTIST AND ENGINEER



SCIENTIST

- Studies how things work in nature.
- Performs experiments and research.
- Discovers new facts and theories.
- Solves scientific questions.
- Uses science and mathematics

ENGINEER

- Applies scientific knowledge to practical problems.
- Designs and develops products or systems
- Builds and tests solutions.
- Improves existing technology.
- Uses science, mathematics, and engineering principles.

Need and Wants of CSE

It generally works by:

1	2
Understanding a need What problem needs solving	Designing a solution Using math, physics, and materials knowledge
3	4
Testing and building Prototyping, analyzing, refining	Optimizing For cost, safety, efficiency, and durability

Needs (why the world requires CSE today)

Digital Infrastructure Banking, healthcare, government, and communication all run on software and networks that need to be built and maintained	Automation & AI Industries need systems that reduce manual labor and improve efficiency (manufacturing, logistics, agriculture)
Cybersecurity As more systems go online, protecting data and infrastructure from attacks is critical	Data-Driven Decisions Businesses and governments rely on data analysis for everything from policy to marketing
Healthcare Tech Diagnostics, medical devices, health records, and AI-assisted treatment all depend on CSE	Smart Devices & IoT Homes, cities, and vehicles increasingly rely on connected, intelligent systems
❏ Computer Science and Engineering (CSE) sits at the center of almost every modern industry today	

DISCIPLINES OF ENGINEERING



- **CIVIL ENGINEERING**

Designs and constructs roads, bridges, buildings, dams, airports, and water supply systems.

- **MECHANICAL ENGINEERING**

Designs, manufactures, and maintains machines, engines, vehicles, and industrial equipment.

- **ELECTRICAL ENGINEERING**

Focuses on electricity, power generation, electronics, telecommunications, and control systems.

- **COMPUTER ENGINEERING**

Combines electrical engineering and computer science to develop computer hardware, embedded systems, and networking devices.

- **CHEMICAL ENGINEERING**

Converts raw materials into useful products such as fuels, pharmaceuticals, food products, and chemicals.

- **AEROSPACE ENGINEERING**

Designs and develops aircraft, spacecraft, satellites, and missiles.



SMART ELECTRIC VEHICLE



Smart Electric Vehicle

Here's a broader picture — CSE doesn't just pair with one department at a time, it actively contributes across many simultaneously in large modern projects. Two ways to see this:

1. One big product touching multiple engineering departments at once

Take a Smart Electric Vehicle (like Tesla) as an example — CSE contributes to almost every department inside a single product:

Department Involved	CSE's Contribution
Mechanical Engineering	Simulation software for chassis/body design, motor control algorithms
Electrical Engineering	Battery management software, power distribution control systems
Automobile Engineering	Autopilot/self-driving software, infotainment

CSE Across Engineering Departments

Department Involved	CSE's Contribution
Civil Engineering	Software for charging station network planning, smart grid integration
Industrial Engineering	AI-driven factory automation, robotics for assembly lines
Environmental Engineering	Software modeling emissions reduction, battery recycling data systems

One product — many departments — CSE is the thread connecting all of them.

2. CSE as a "service" discipline contributing skills to many departments

Skill CSE Brings	Departments It Helps
AI/Machine Learning	Biomedical (diagnostics), Civil (smart cities), Agricultural (crop prediction)
Embedded Systems Programming	Electrical, Mechatronics, Aerospace (avionics)
Data Analytics	Industrial (supply chain), Environmental (climate data), Chemical (process optimization)
Simulation Software	Mechanical (CAD/CFD), Structural (load analysis), Aerospace (flight sim)
Robotics & Automation	Mechanical, Industrial, Agricultural
Cybersecurity	Electrical (smart grids), Civil (infrastructure control systems)

□ The bigger idea: CSE isn't tied to just one engineering

Misconception of CSE

Here are some simple points on Misconceptions of CSE (Computer Science and Engineering):

CSE is only about coding. ✖

Only very intelligent students can study CSE. ✖

CSE students only use computers all day. ✖

CSE graduates can only become software engineers. ✖

Mathematics is the only important subject in CSE. ✖

CSE is easy because it only involves typing code. ✖

Girls cannot succeed in CSE. ✖

You do not need communication skills in CSE. ✖

☐ Reality: CSE is a broad field that includes

SKILLS REQUIRED FOR CSE IN 21ST CENTURY



- **PROGRAMMING SKILLS** – Learn languages like C, Python, Java, or C++.
- **PROBLEM-SOLVING SKILLS** – Find and solve technical problems efficiently.
- **LOGICAL THINKING** – Think step by step to write correct programs.
- **COMMUNICATION SKILLS** – Explain ideas clearly and work with others.
- **TEAMWORK** – Collaborate effectively in group projects.
- **CREATIVITY** – Develop new ideas and innovative solutions.
- **CRITICAL THINKING** – Analyze problems before making decisions.
- **ADAPTABILITY** – Learn new technologies and keep up with changes



ABOUT GAMMA AI

Gamma is an AI-powered tool that helps users create presentations, documents, webpages, and reports quickly using simple text prompts. Instead of designing slides manually, you describe your topic, and Gamma generates a professional-looking presentation with content, layouts, and images.

FEATURES OF GAMMA AI

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THANK YOU



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